EPS Rev B — integrated development schematic

Initial checkpoint: controller + CAN only. Remaining power, BMS, charger and protection sheets pending.

Controller

File: controller.kicad_sch

CAN interfaces

File: can_interfaces.kicad_sch

Reference ranges: 100 controller, 200 BMS, 300 charger, 400 essential 3V3, 500 recovery 5V, 600 protection.

External interface names preserved. Logical connectors are placeholders; no production connector acceptance.

Incomplete ERC remains visible. No PCB synchronization or fabrication acceptance.

S03 severity API unavailable; S11 required field display unsupported; reused custom symbol S08 review pending.

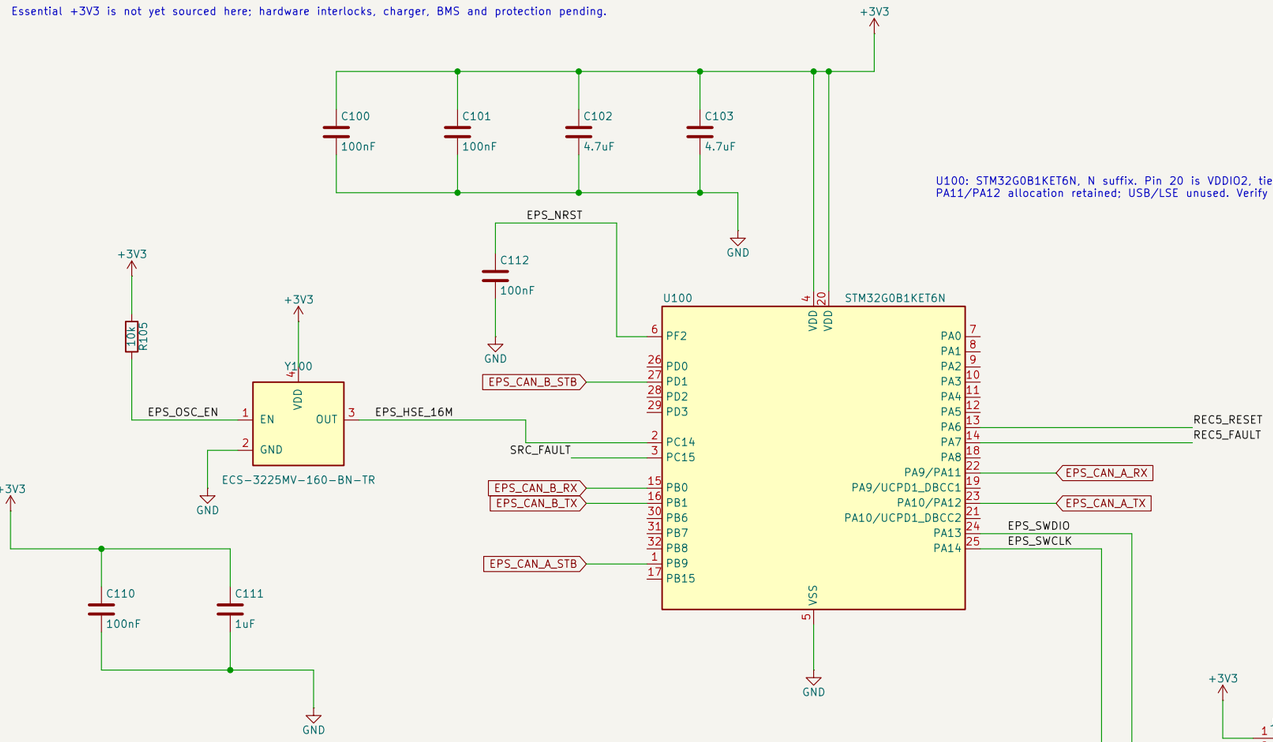
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CONTROLLER — integrated development checkpoint

Essential +3V3 is not yet sourced here; hardware interlocks, charger, BMS and protection pending.



U100: STM32G0B1KET6N, N suffix. Pin 20 is VDDIO2, tied to VDD.
PA11/PA12 allocation retained; USB/LSE unused. Verify UCPD reset configuration.

PROTECTION INTERFACES — NOT YET LINKED
PA6 REC5_RESET -> existing 47k Q107 base path in protection study.
PA7 REC5_FAULT; PC15 SRC_FAULT: active-low, 47k essential pullups there.
Reset: 2 ms pulse, >=10 s cooldown and rail qualification.
Reset is not permission to bypass protection.

J100: logical SWD access; physical connector unselected.
1 VTref sense only, 2 SWDIO, 3 SWCLK, 4 NRST, 5 GND.
No injected probe power. Power-off/backfeed review open.

16 MHz HSE bypass; no passive crystal.
Place C110/C111 locally at Y100 in layout; source-series option open.

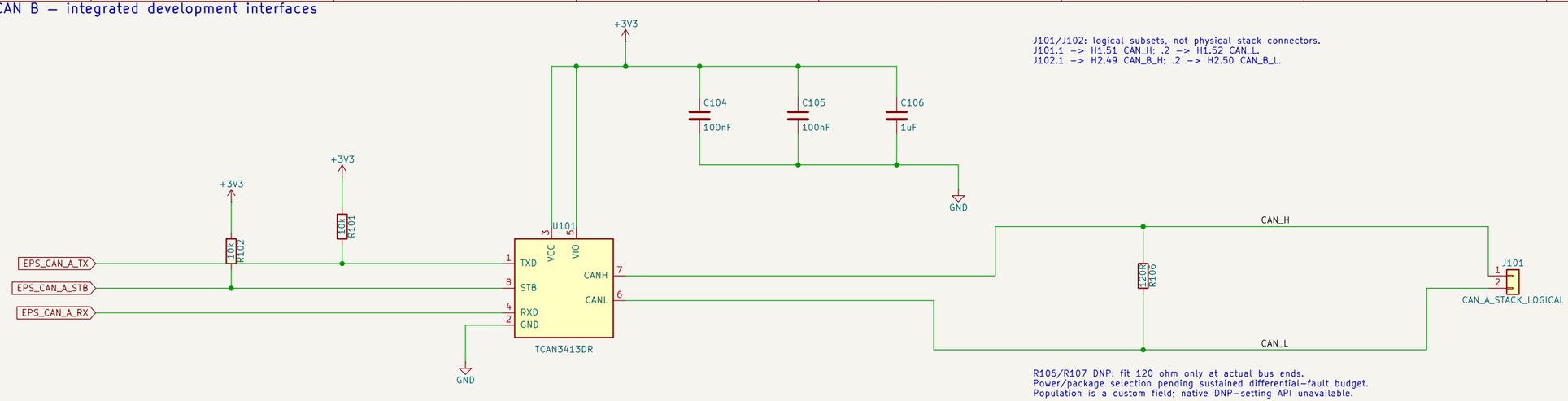
OPEN: ADC conditioning, local charger I2C, isolated pack SMBus,
UART, interlock status and payload request. Required pins remain unconnected.
No EPS drive on IHU-owned COMMS_EN / PAYLOAD_EN or stack alert.

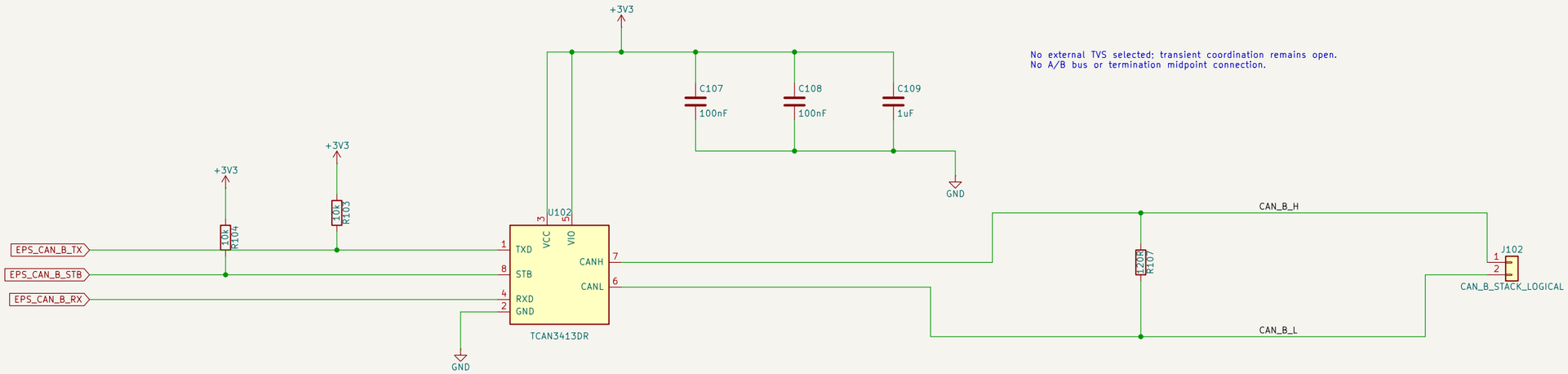
PD3 reserved. Do not connect BMS BTP directly to 3V3 logic.
Unused BTP is tied to BMS VSS in the proposed BMS circuit; poll pack status.

S11 OPEN: metadata is saved; required visible field styling unsupported.
S08: reused TCAN/ECS custom symbols remain under original construction review.

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CAN A / CAN B — integrated development interfaces





FAULT LIMIT: shared +3V3, GND, MCU and clock remain common paths.
Independent VCC/VIO + pullup branch limiting remains a written candidate.
CAN B is bus-path recovery, not independent controller/power redundancy.

Existing TCAN custom symbols and manufacturer patterns reused.
Pin maps verified; assembly, ESD, timing and physical placement not qualified.
S08 construction review / S11 field-display limitations remain open.

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